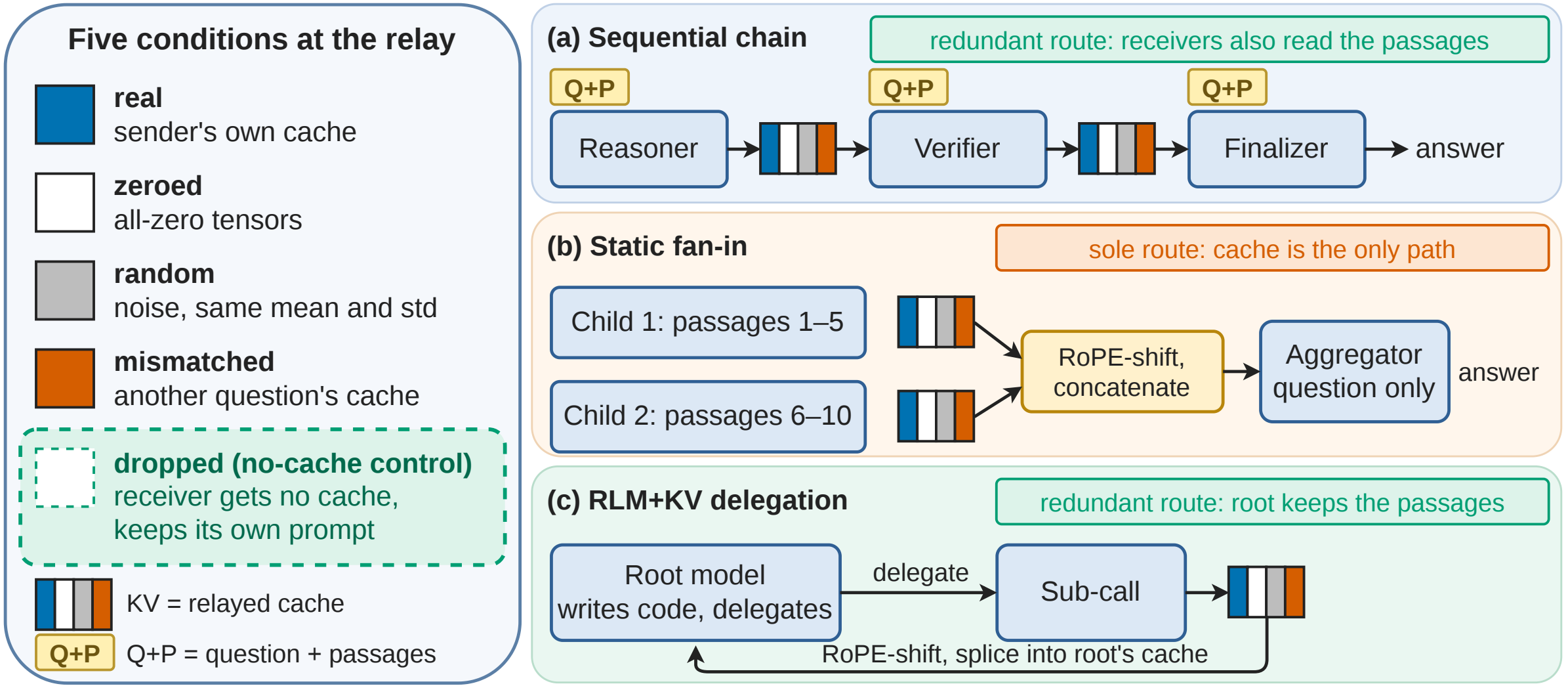
Five conditions at the relay
real
sender's own cache
zeroed
all-zero tensors
random
noise, same mean and std
mismatched
another question's cache
dropped (no-cache control)
receiver gets no cache,
keeps its own prompt
KV = relayed cache
Q+P
Q+P = question + passages
(a) Sequential chain
redundant route: receivers also read the passages
Q+P
Q+P
Q+P
Reasoner
Verifier
Finalizer
answer
(b) Static fan-in
sole route: cache is the only path
Child 1: passages 1–5
Child 2: passages 6–10
RoPE-shift,
concatenate
Aggregator
question only
answer
(c) RLM+KV delegation
redundant route: root keeps the passages
Root model
writes code, delegates
delegate
Sub-call
RoPE-shift, splice into root's cache